

Geography Ch-5 (Must-Do) — Landforms Made by Running Water (Rivers)

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain

River work: erosion, transport, deposition through upper / middle / lower course

CA anchor: Ganga-Brahmaputra delta erosion (sediment starvation)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — The Work of a River: Upper, Middle & Lower Course (GC Leong)

GS-I • Physical
Geography →
Fluvial Landforms
(Source: GC
Leong)

■ NEWS TRIGGER

River geomorphology underpins India's water security and disaster profile — from Himalayan gorges to the deltas now eroding due to upstream dams cutting sediment supply.

■ INTRO & ORIGIN

A river erodes, transports and deposits. Its work changes down the long profile: **vertical corrasion** dominates the upper course, **lateral corrasion** the middle course, and **deposition** the lower course.

Erosion mechanisms (GC Leong): **corrasion** (mechanical wear — vertical deepens, lateral widens), **corrosion/solution** (chemical, e.g. limestone dissolving), **hydraulic action** (water force in cracks) and **attrition** (load fragments wearing each other).

■ KEY DATA

Course	Dominant process	Characteristic landforms
Upper (mountain)	Vertical corrasion	V-shaped valley, gorge, rapids, waterfalls, interlocking spurs
Middle (plain edge)	Lateral corrasion + transport	Meanders, river cliffs, valley widening, river capture
Lower (flood plain)	Deposition	Flood plains, ox-bow lakes, natural levees, braided channels, delta

■ STATIC THEORY

- **Upper course:** swift descent, rapid down-cutting → deep, narrow **V-shaped valley**; hard-rock bands give **rapids and waterfalls**; spurs interlock.
- **Middle course:** gradient eases; vertical corrasion nearly ceases, **lateral corrasion** widens the valley; the river swings in **meanders** and cuts **river cliffs**.
- **Lower course:** heavy with debris, work is mainly **deposition** — building **flood plains**; cut-off meanders become **ox-bow lakes**; sediment builds **natural levees**.
- At the mouth, sediment forms a **delta** (arcuate/bird's-foot/estuarine) or, where tides scour it away, an **estuary**. Majid Husain lists lower-valley forms: **braided channel, delta, estuary, alluvial deposits**.
- The **graded long profile** is the smooth, concave-up curve a river tends toward from source to mouth.

■ MUST-KNOW FACTS

1. Corrasion = mechanical erosion: vertical (deepens) vs lateral (widens).
2. Corrosion/solution = chemical erosion (limestone dissolves).
3. Upper course landform: V-shaped valley, gorge, waterfall, interlocking spurs.
4. Middle course: meanders + river cliffs (lateral corrasion).

■ UPSC TRAPS

- ✗ V-shaped valleys and U-shaped valleys are both made by rivers.
- ✓ Rivers cut V-shaped valleys; U-shaped valleys are glacial.
- ✗ Deposition dominates the upper course.
- ✓ Vertical corrasion (erosion) dominates the upper course; deposition dominates the LOWER course.

5. Lower course: flood plain, ox-bow lake, levee, delta / braided channel.

6. Estuary forms where tides remove sediment instead of building a delta.

✗ Every river ends in a delta.

✓ Where strong tides scour sediment, an ESTUARY forms instead (e.g. Narmada/Tapi).

■ MAINS ANGLE

Trace how a river's dominant process shifts from erosion to deposition along its course, and relate this to India's Himalayan gorges, Gangetic flood plains and coastal deltas.

■ STUDY LINK

Geography → Physical → Fluvial Landforms (GC Leong: Ch on Running Water; Majid: River Landforms)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — Ganga-Brahmaputra Delta Erosion & Sediment Starvation

GS-I / GS-III ·
Fluvial deposition
applied → Delta
management

■ NEWS TRIGGER

The Ganga-Brahmaputra (Sundarbans) delta is eroding as upstream dams and barrages cut the sediment supply that normally builds and maintains the delta — worsened by sea-level rise and mangrove loss.

■ INTRO & ORIGIN

A delta survives only if the sediment deposited at the mouth outpaces erosion and subsidence. 'Sediment starvation' from upstream trapping tips the balance toward net land loss.

■ STATIC THEORY

- Deltas are the classic **depositional** landform of the lower course — sediment starvation reverses the delta-building process.
- Causes: upstream **dams/barrages trap sediment**, **sea-level rise**, embankments, and **mangrove depletion** (Sundarbans).
- Impacts: loss of fertile land, saltwater intrusion, threats to mangrove biodiversity and migratory-bird habitat.

■ MUST-KNOW FACTS

1. Delta = depositional landform; needs sediment supply > erosion.
2. Key term: 'sediment starvation' from upstream dams/barrages.
3. Ganga-Brahmaputra delta = Sundarbans (largest delta / mangrove).

■ UPSC TRAPS

✗ Dams always protect downstream deltas.

✓ Dams TRAP sediment upstream, starving and shrinking the delta.

■ MAINS ANGLE

Link fluvial deposition theory to delta erosion, arguing for integrated basin/sediment management.

■ STUDY LINK

Geography → Fluvial Deposition → Deltas (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI